Compliance Summary with Evaluation and Segmentation Guidelines

Project: Radar–Based Kick Gesture Recognition using CNNs
Student: *Aman Syed*
Date: 18- Dec-2025

This section outlines how the supervisor's standards and methodological guidelines are closely adhered to in the proposed project design, implementation, and evaluation process. This section is included to make it evident that all important instructions have been fully implemented in the codebase and the experimental analysis, particularly those pertaining to data segmentation, subject-wise evaluation, and sliding-window validation.

Alignment with Project Objective

The project's main goal is to recognize kick gestures using deep learning methods and mmWave radar emissions. Designing a reliable, person-independent gesture recognition pipeline and methodically comparing several segmentation techniques under the same training and assessment conditions are the main objectives. This is in line with the supervisor's advice to assess segmentation techniques rather than architectural differences.

Dataset Preparation and Preprocessing

A fixed parent window of 12000 samples, recorded at 4000 Hz, is used to standardize all radar recordings. This guarantees that every gesture instance has the same temporal context. Band-pass filtering (50–1999 Hz) is used to preprocess the raw I/Q radar signals in order to eliminate noise and DC components. All channels are then scaled into the range [−1, 1] using a global min–max normalization, which stabilizes CNN training and guarantees comparability between subjects and sessions.

Strict subject-wise data division is made possible by the clear metadata associated with each recording, such as Person ID, Gesture Class, and Repetition Index.

Person-Wise Train–Test Splitting

The dataset is subjected to a rigorous 80/20 split per person. Training and validation sets do not include any subjects from the test set. Programmatic enforcement of group-based separation prevents subject leaking. In contrast to memorization of specific motion patterns, this guarantees that model performance reflects real generalization to unseen users.

Fixed-Length Training Strategy

From each 12000-sample parent window, a **single fixed-length segment** is extracted for training. Three window lengths are evaluated:

- 4000 samples
- 2000 samples
- 1000 samples

Compliance Summary with Evaluation and Segmentation Guidelines

Separate CNN models are trained for each window length. This design allows direct analysis of how temporal context influences recognition performance while maintaining consistent training conditions.

Sliding-Window Evaluation on Unseen Data

A sliding-window inference approach is used to evaluate performance for test subjects who have not yet been seen. Overlapping windows of length 4000, 2000, or 1000 samples are slid across the entire 12000-sample test signal rather than evaluating on a single extracted segment. Each window's predictions are produced by the model, and the ultimate choice is made by adding the probabilities from each window.

The supervisor's demand to evaluate using sliding-window analysis on unseen data is directly satisfied by this evaluation methodology, which closely resembles real-world deployment settings where the exact timing of the kick gesture is unknown.

Segmentation Methods Compared

Under the same circumstances, three segmentation procedures are used and assessed:

1. Short-time signal energy is the basis for energy peak segmentation.
2. Gradient Peak Segmentation: based on the signal magnitude's temporal gradient
3. Time-frequency-based Doppler (STFT) Peak Segmentation Doppler concentration of energy

The CNN design, preprocessing pipeline, and assessment protocol are the same for all segmentation techniques. This guarantees that architectural bias is not the cause of performance disparities, but rather the segmentation technique.

Model Architecture and Training Control

Every experiment uses the same 1D convolutional neural network design. To avoid overfitting, early halting based on validation loss is used. Loss and accuracy curves are used to depict training dynamics, giving insight into the model's convergence behavior.

Evaluation Metrics and Reporting

Fixed-length training windows, person-wise data splitting, sliding-window evaluation on unseen subjects, and structured technique comparison are among the supervisor-specified needs that have all been properly implemented and tested. As a result, the experimental process is replicable, methodologically sound, and in line with both academic best practices and practical deployment considerations.

Final Statement of Compliance

All specified requirements - fixed-length training windows, person-wise data splitting, sliding-window evaluation on unseen subjects, and structured method comparison—have been fully implemented and validated. The experimental pipeline is therefore methodologically sound, reproducible, and aligned with both academic best practices and real-world deployment considerations.